M&AE 5070 - Flight Dynamics

Chapter 9 - Lateral Dynamics

Companion Notes to Case Study 7

The study of lateral dynamics will be follow closely what was done for the longitudinal dynamics, and that starts with the linearization of the equations.

1 Linearized lateral equations

The reference equilibrium state is straight, level flight with no side-slip. This implies horizontal wings and no rotation velocity, that is, banking angle, side-slip, and yaw, roll and pitch velocities all equal to zero:

$$\text{Lateral equilibrium state: } \phi_e = \beta_e = r_e = p_e = q_e = 0$$

The initial lateral state variables are therefore all zero when stability is considered. This was not the case for the longitudinal dynamics, where α, γ, and altitude could be non-zero. It means that instead of working with variable *changes* ($\overline{\alpha} = \alpha - \alpha_e$, $\overline{\gamma} = \gamma - \gamma_e$ etc.), we will work directly with the variables themselves. To simplify the algebra, we also assume that the product of inertia E (one of the non-diagonal term in matrix of inertia) is zero, and as usual, any side slip angle remains small.

The objective is to obtain a linear system of equations relating the lateral parameters β, p, r, and ϕ to their temporal derivatives $\dot{}$:

$$\text{State Vector } \begin{pmatrix} \beta \\ p \\ r \\ \phi \end{pmatrix} \qquad \begin{pmatrix} \dot{\beta} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \end{pmatrix} = \underline{\underline{A}} \begin{pmatrix} \beta \\ p \\ r \\ \phi \end{pmatrix} + \underline{\underline{B}} \begin{pmatrix} \delta_\ell \\ \delta_n \end{pmatrix}$$

$\underline{\underline{A}}$: 4 × 4 constant matrix evaluated at starting equil.

$\underline{\underline{B}}$: 4 × 2 constant matrix, effects of controls.

Example: Linearized LF Lateral Force equation (first line, equation for $\dot{\beta}$)

$$mV\left(\dot{\beta} + r\cos\alpha_e - p\sin\alpha_e\right) = mg\sin\phi\cos\theta_e + \frac{1}{2}\rho S V_a^2 C_Y \tag{1}$$

$$\text{with } C_Y = C_{Y_\beta}\beta + C_{Y_{\delta_\ell}}\delta_\ell + C_{Y_{\delta_n}}\delta_n \tag{2}$$

Extracting $\dot{\beta} = d\beta/dt$ out of the equations, we get:

$$\dot{\beta} = \underbrace{\frac{mg}{mV}\sin\phi\cos\theta_e + \frac{1}{2mV}\rho S V^2 C_Y - r\cos\alpha_e + p\sin\alpha_e}_{\text{RHS}} \tag{3}$$

The linearized version of the equation above is obtained by taking the multi-variate, first order Taylor expansion of it:

$$\dot{\beta} = \frac{\partial \text{RHS}}{\partial\beta}\beta + \frac{\partial \text{RHS}}{\partial r}r + \frac{\partial \text{RHS}}{\partial p}p + \frac{\partial \text{RHS}}{\partial\phi}\phi + \frac{\partial \text{RHS}}{\partial\delta_\ell}\delta_\ell + \frac{\partial \text{RHS}}{\partial\delta_n}\delta_n \tag{4}$$

For example:

$$A_{11} = \frac{\partial\dot{\beta}}{\partial\beta} = \frac{\partial\,\text{RHS}}{\partial\beta}\bigg|_{r,p,\phi,\delta_\ell,\delta_n} = \frac{\partial}{\partial\beta}\left(\frac{1}{2mV}\rho S V^2 (C_{Y_\beta}\beta)\right) = \frac{1}{2mV}\rho S V^2 C_{Y_\beta}$$

The complete set of linearized equations are:

$$\begin{pmatrix}\dot\beta\\ \dot r\\ \dot p\\ \dot\phi\end{pmatrix} = \begin{pmatrix}\frac{Y_\beta}{V} & -\cos\alpha & \sin\alpha & \frac{g\cos\theta}{V}\\ n_\beta & n_r & n_p & 0\\ \ell_\beta & \ell_r & \ell_p & 0\\ 0 & \tan\theta & 1 & 0\end{pmatrix}\begin{pmatrix}\beta\\ r\\ p\\ \phi\end{pmatrix} + \begin{pmatrix}\frac{Y_{\delta_\ell}}{V} & \frac{Y_{\delta_n}}{V}\\ n_{\delta_\ell} & n_{\delta_n}\\ \ell_{\delta_\ell} & \ell_{\delta_n}\\ 0 & 0\end{pmatrix}\begin{pmatrix}\delta_\ell\\ \delta_n\end{pmatrix} \tag{5}$$

The various terms in these matrices are:

$$Y_\beta = \tfrac{1}{2m}\rho S V^2 C_{Y_\beta} \qquad Y_{\delta_\ell} = \tfrac{1}{2m}\rho S V^2 C_{Y_{\delta_\ell}} \qquad Y_{\delta_n} = \tfrac{1}{2m}\rho S V^2 C_{Y_{\delta_n}}$$

$$n_\beta = \tfrac{1}{2C}\rho S\ell V^2 C_{n_\beta} \qquad n_p = \tfrac{\ell}{2CV}\rho S\ell V^2 C_{n_p} \qquad n_r = \tfrac{\ell}{2CV}\rho S\ell V^2 C_{n_r}$$

$$n_{\delta_\ell} = \tfrac{1}{2C}\rho S\ell V^2 C_{n_{\delta_\ell}} \qquad n_{\delta_n} = \tfrac{1}{2C}\rho S\ell V^2 C_{n_{\delta_n}}$$

$$\ell_\beta = \tfrac{1}{2A}\rho S\ell V^2 C_{\ell_\beta} \qquad \ell_p = \tfrac{\ell}{2AV}\rho S\ell V^2 C_{\ell_p} \qquad \ell_r = \tfrac{\ell}{2AV}\rho S\ell V^2 C_{\ell_r}$$

$$\ell_{\delta_\ell} = \tfrac{1}{2A}\rho S\ell V^2 C_{\ell_{\delta_\ell}} \qquad \ell_{\delta_n} = \tfrac{1}{2A}\rho S\ell V^2 C_{\ell_{\delta_n}}$$

We can also rewrite all the terms in a more compact form by using **S** :

$A_{11} = \frac{1}{2mV}\rho S V^2 C_{Y_\beta} = \frac{Y_\beta}{V}$ **S** $\frac{1}{2}\rho S V^2 = \frac{mg}{C_z} \rightarrow Y_\beta = \frac{mg}{mC_z} C_{Y_\beta} = \frac{g C_{Y_\beta}}{C_z}$

The four equations of the state matrix lead to three modes:

* 2 aperiodic modes : ROLL mode (state variable : p)
  SPIRAL mode (state variable : ϕ)

* 1 oscillatory mode : SIDE SLIP oscillations, DUTCH ROLL
  (state variables : β, r)

Using decoupling arguments, we can again individually study each of them analytically.

Validity of linearized equations. In contrast to the longitudinal motion, it is common to have bamnk angles reach 30°, even for commercial, transport planes! In these cases, the linearization around the horizontal aircraft equilibrium is questionable, especially for the slow modes. The answer is to linearize around the steady turn state, with $\phi \neq 0$, which in theory, does not cause any difficulty, but certainly complicates the linearized equations.

Additional simplifying assumptions. To simplify even further, we'll assume that

- θ and α are zero → non-zero from longitudinal equations, but approximation ok when just looking at lateral
- the lateral force due to the roll control δ_ℓ is zero: $C_{Y_{\delta_\ell}} = 0$
- (sometimes) the roll/yaw coupling term is zero: $C_{n_p} \approx 0$ (see later)

Then, the linearized system become:

LF Y R KΦ

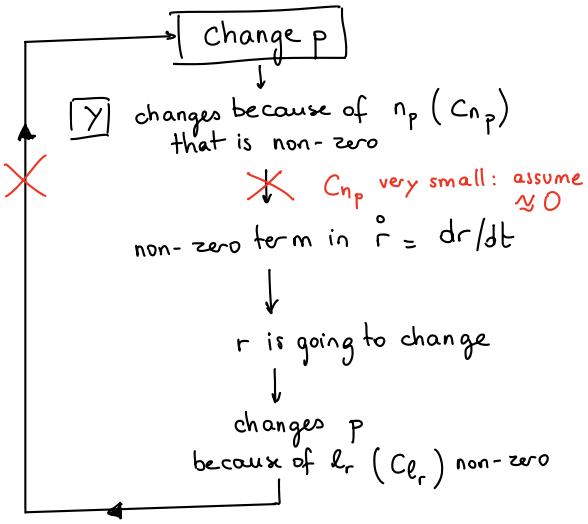
$$\begin{pmatrix}\dot\beta\\ \dot r\\ \dot p\\ \dot\phi\end{pmatrix} = \begin{pmatrix}\frac{Y_\beta}{V} & -1 & 0 & \frac{g}{V}\\ n_\beta & n_r & n_p & 0\\ \ell_\beta & \ell_r & \ell_p & 0\\ 0 & 0 & 1 & 0\end{pmatrix}\begin{pmatrix}\beta\\ r\\ p\\ \phi\end{pmatrix} + \begin{pmatrix}0 & \frac{Y_{\delta_n}}{V}\\ n_{\delta_\ell} & n_{\delta_n}\\ \ell_{\delta_\ell} & \ell_{\delta_n}\\ 0 & 0\end{pmatrix}\begin{pmatrix}\delta_\ell\\ \delta_n\end{pmatrix} \tag{6}$$

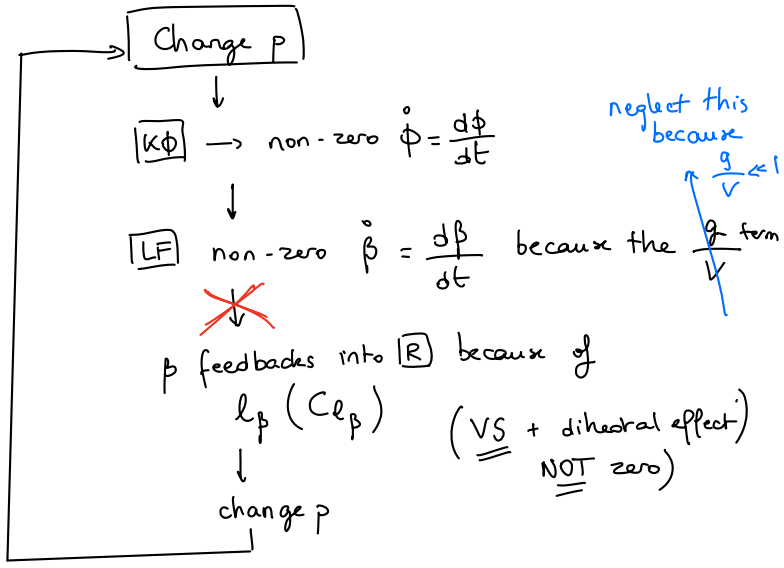
2 Roll mode

Experience shows that this mode is strongly associated with the Roll moment equation **R** . To study it, we need to decouple it from the other three state variables, that is, we need to show that:

When the roll velocity is "active" ($p \neq 0$), it does not create anything on the other state variables/equations that would feed back into R

$$\begin{matrix} \text{LF} \\ \text{Y} \\ \text{R} \\ \text{K}\phi \end{matrix} \begin{pmatrix} \dot\beta \\ \dot r \\ \dot p \\ \dot\phi \end{pmatrix} = \begin{pmatrix} \frac{Y_\beta}{V} & -1 & 0 & \frac{g}{V} \\ n_\beta & n_r & n_p & 0 \\ \ell_\beta & \ell_r & \ell_p & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \beta \\ r \\ p \\ \phi \end{pmatrix} + \begin{pmatrix} 0 & \frac{Y_{\delta_n}}{V} \\ n_{\delta_\ell} & n_{\delta_n} \\ \ell_{\delta_\ell} & \ell_{\delta_n} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \delta_\ell \\ \delta_n \end{pmatrix}$$





Decoupling assumptions for Roll mode.

- Yaw / Roll coupling negligible $C_{n_p} \approx 0$
- component of weight on lateral force is negligible because $\frac{g}{V} \ll 1$

Interpretation. Those decoupling assumptions are equivalent to saying that a non-zero roll velocity does not excite **LF** and **Y** , but a non-zero yaw velocity or side-slip angle do excite the roll equation.

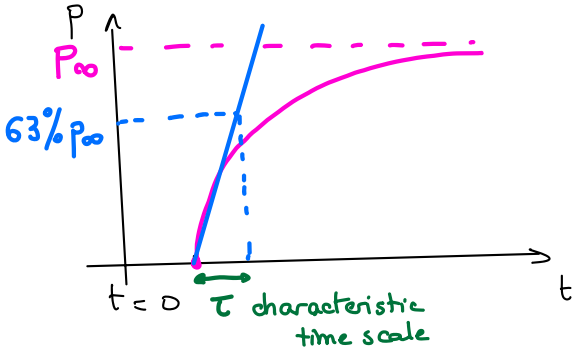
Decoupled linearized equation of Roll mode

$$\dot p - \ell_p\, p = \ell_{\delta_\ell}\, \delta_\ell + \left[\ell_\beta\, \beta + \ell_r\, r \right]$$

We want to investigate the response of our aircraft to a deflection of the ailerons $\Delta\delta_\ell$. The solution of the equation above is:

activate roll mode using ailerons

$$p = -\frac{\ell_{\delta_\ell}}{\ell_p}\left(1 - e^{\ell_p t}\right)\delta_\ell$$



Characteristic time τ

moment of inertia around $\vec{x}_b$ $(A>0)$

$$\tau = -\frac{1}{\ell_p} = -\frac{2A}{\rho S V \ell^2 C_{\ell_p}}$$

$C_{\ell_p} < 0$ (stabilizing effect)

τ is a function of V, A, C_{ℓ_p}

but independent of $C_{\ell_{\delta_\ell}}$

Order of magnitude $0.5\ s < \tau_{95\%} < 1.4\ s$

δ_ℓ "directly controls roll velocity".

Final roll velocity p_∞

3 Side-slip oscillation or Dutch Roll

This mode is an oscillation between the side slip angle β and the yaw velocity r. We will see that this oscillation creates an important roll motion as well during flight, hence the "Dutch Roll" name (the "Dutch" part has various more or less serious explanations, which wikipedia will be happy to provide).

Experience shows that this mode is associated with the **LF** and **Y** equations. The decoupling assumptions are the same as those for the Roll mode. Indeed, if β and r create some roll velocity p through the aerodynamic contributions C_{ℓ_β} and C_{ℓ_r}, this roll velocity does not feed back to **LF** and **Y** for the same reasons as mentioned above: the corresponding two lines in the linearized matrix equations are indeed decoupled from the rest.

Decoupled linearized equations for Dutch roll.

$$\begin{pmatrix} \dot{\beta} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \frac{Y_\beta}{V} & -1 \\ n_\beta & n_r \end{pmatrix} \begin{pmatrix} \beta \\ r \end{pmatrix} + \begin{pmatrix} \frac{Y_{\delta_\ell}}{V} & \frac{Y_{\delta_n}}{V} \\ n_{\delta_\ell} & n_{\delta_n} \end{pmatrix} \begin{pmatrix} \delta_\ell \\ \delta_n \end{pmatrix} \tag{7}$$

On top of that, a roll motion will be created, governed by the Roll mode equation:



Figure 1: The B2 bomber, a flying wing. This configuration leads to very low dutch roll damping ratios. However, there is no instability, because a flying wing (no vertical stabilizer) has a $C_{n_\beta} > 0$.

3.1 Natural characteristics of Dutch Roll

Canonical form of characteristic polynomial (damped harmonic oscillator):

We need to compare this canonical form to the equation yielding the eigenvalue of the matrix:

In practice, $n_r \frac{Y_\beta}{V} \ll n_\beta$

- **Natural frequency**:

- **Damping**:

- **Damping ratio**:

Average values of ω_n are around 2 rad/s (or a period of $\approx$ 3 sec), and damping ratios are around 0.1. Regulations imposes

Parallels with short-mode oscillations There is a very strong analogy between the Dutch roll and the short-period mode for longitudinal flight.

Consequences of a low damping ratio This mode causes more problems in terms of flight handling qualities than its longitudinal counterpart, because it is less damped than the rapid-period mode (damping is limited by the size of the vertical stabilizer). It can also unfortunately get coupled with the the roll moment. If we investigate the relative importance of both terms inside parenthesis for ζ:

$$\frac{l^2}{C} C_{n_r} + \frac{C_{y\beta}}{m}$$

C : moment of inertia can be defined as

$$C = m \mathcal{R}_{yz}^2$$

$$\frac{d}{\mathcal{R}_{yz}} \sim 1.4 - 1.8$$

damping coming from the wings, a bit from the fuselage, but mostly from the vertical stabilizer

$$C_{n_r} \approx \left(\frac{1}{k_{n_r}} \frac{d}{l} \right)^2 C_{y_\beta}$$

order of magnitude :

$$\frac{1}{k_{n_r}} \frac{d}{l} \approx 2$$

$$\Rightarrow \frac{\ell^2}{c} C_{n_r} + \frac{C_{Y_\beta}}{m} = \frac{C_{Y_\beta}}{m}\left(\frac{1}{k_{n_r}^2}\frac{\ell^2}{R_{yz}^2}\frac{d^2}{\ell^2} + 1\right)$$

$$= \frac{C_{Y_\beta}}{m}\left(\frac{d^2}{k_{n_r}^2 R_{yz}^2} + 1\right)$$

part due to C_{n_r} — part due to C_{Y_β}

→ Can be of order 2

- The damping ratio for the short period mode had a similar form, however, both terms in the sum for the short period mode were roughly the same in magnitude.
- In the case of the dutch roll, the C_{n_r} contribution can be twice as large as the C_{Y_β} contribution.

This has a clear impact on the value of the damping ratio:

How can we "pilot" a Dutch roll, that is, act on controls to damp its oscillations faster?

4 Spiral Mode

The last mode is different from the other ones in that it is the only mode allowed to be "slightly" unstable. From the physical point of view, we consider the situation where the aircraft is turning. The pilot is setting all controls to neutral ($\delta_\ell = \delta_n = 0$)

- The spiral mode is **stable**:

- The spiral mode is **unstable**:

This mode is aperiodic (associated with the kinematic equation $\dot{\phi} = p$, and it is very slow. We can obtain good analytical solution by setting all equations to equilibrium, except for the one for $\dot{\phi} = d\phi/dt$.

- If $\theta = \alpha = 0$: too much simplification, the eigenvalue associated with the mode is 0, which means the mode is infinitely slow. Correct trend, but not useful!

- Instead, we keep θ and α in the equations and assume they are small. We then get the following set of equations:

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} \frac{Y_\beta}{V} & -\cos\alpha & 0 & \frac{g}{V} \\ n_\beta & n_r & n_p & 0 \\ \ell_\beta & \ell_r & \ell_p & 0 \\ 0 & \tan\theta & 1 & 0 \end{pmatrix} \begin{pmatrix} \beta \\ r \\ p \\ \phi \end{pmatrix} \tag{8}$$

This system of equations can be solved analytically for the time constant of the spiral mode.

Only the coefficient linked to the dihedral effect can change signs, by changing how the wing is fixed on the fuselage. The other coefficients are always of the same signs, regardless of the aircraft:

To get a stable spiral, C_{ℓ_β} should at least be negative (wing with a positive dihedral, or a sufficiently large sweep), since $C_{\ell_r} C_{n_\beta}$ is always positive. We can extract a rule of thumb for how high the dihedral angle Γ needs to be:

The geometric ratio $\frac{d}{\ell} \approx 3$, and the order of magnitude for the k coefficients are $k_{n_\beta} \approx 3$ and $k_{n_r} \approx \frac{3}{2}$. We established models for C_{ℓ_β} and C_{ℓ_r} in a previous chapter. From those models, we can estimate the dihedral angle necessary to ensure a stable spiral (assuming an average taper ratio of $\lambda = 0.3$):

$$\frac{\Gamma}{\bar{\alpha}} + 2\tan\phi > \frac{k_{n_r}^2}{k_{n_\beta}} \frac{b}{d} \left(1 + \frac{\lambda}{1+4\lambda}\right) \approx 0.7 \frac{b}{d} \tag{9}$$

For a cruising transport aircraft, $C_z \approx 0.5$, corresponding to $\bar{\alpha} \approx 6^o$, leading to a dihedral angle of $\Gamma \approx 3^o$to ensure the stability of the spiral mode (for a wing without sweep).

Regulations do not impose the spiral mode to be stable, it is the only mode allowed to be reasonably unstable. In this case, the time needed to double the bank angle starting at 20^oshould be larger than 12 seconds.

How to tune the spiral mode Among the four aerodynamic coefficients that contribute to the stability criterion, only the dihedral effect C_{ℓ_β} can be changed by changing how the wings are mounted on the fuselage:

- C_{ℓ_r} is function of the shape of the wing, which is more or less fixed by the longitudinal performance constraints.
- C_{n_r} and C_{n_β} are function of the size and location of the vertical stabilizer, which are fixed by the lateral performances, and constrained by engine failure considerations. Those coefficients also play a role in the stability of the dutch rollm so we may not want to touch them anyway.

It is therefore reasonable to tune the spiral mode using the dihedral effect C_{ℓ_β}. **If C_{ℓ_β} is increased**, the spiral mode is stabilized. A spiral mode **too stable** will require constant stick action to maintain a turn, which may become annoying for the pilot. On the other hand, an increased C_{ℓ_β} increases the contribution of roll to the dutch roll, which again, is not desirable from the flight handling qualities point of view. However, the aircraft will come back to a neutral (horizontal) position on its own if the stick is set back to neutral, which is convenient and reassuring.

For a modern aircraft with electric controls that take care of flight safety, it is possible to favor pilot's comfort by tuning the spiral mode to be neutral: once the aircraft starts a turn, pilot can leave the stick in a neutral position, the bank angle will be maintained (unless this bank angle is large, in which case the aircraft becomes stable again, and the pilot needs to maintain the stick on one side to keep the aircraft in its turn.